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RESEARCH PAPER

Understanding decentralized autonomous organizations from the inside

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Abstract
Blockchain technology is argued to drastically change the way we operate within an organizational context, with decentralized autonomous organizations (DAOs) representing a first manifestation of this ongoing trend. DAOs are characterized by an online community that builds the organization's backbone by providing knowledge and human resources in a transparent, virtual manner, as well as the use of blockchain technology to coordinate their endeavor. Nevertheless, current research highlights the conceptual ambiguity of this emerging phenomenon, leading to potential issues for practitioners and researchers. To provide further clarity on the phenomenon, we study DAOs through the perspective of their members with a two-staged approach by combining elements of a netnographic approach and structural topic modeling. Our findings highlight several contextual features surrounding DAOs, such as their members' underlying beliefs and views, helping to embed DAOs in existing research streams.

Keywords DAO · Online community · Reddit · Open-Source-Software · Structural topic modeling

JEL Classification 039

Introduction

With the introduction of Bitcoin in 2008 and its tremendous growth with several price peaks over the last decade, distributed ledger technology, often referred to as blockchain, has gained large-scale media attention and academic interest (Kollewe, 2021; Rossi et al., 2019). While in its beginnings, the usage of blockchain technology was mainly limited to cryptocurrencies, this changed in 2014 with the introduction of Ethereum, opening the way for the application of blockchain technology in companies (Buterin, 2014). In the organizational context, blockchain technology is argued to mediate the organizational structure towards a flatter hierarchy with significant implications on various aspects of the organization, such as the organization's performance (Zhao et al., 2022) or the level of autonomy (Beck et al., 2018; Rossi et al., 2019). This essentially offers new organizing approaches that depart entirely from the way we used to work and collaborate (Schirrmacher et al., 2021). One distinct form of such hierarchical flatter forms of organizing is known as decentralized autonomous organizations (DAOs) (Beck et al., 2018; Ziolkowski et al., 2020). DAOs represent the technical instance of an ongoing trend in information systems (IS) research (Beck et al., 2018; Rossi et al., 2019) and practice towards more decentralized organizations (Lee & Edmondson, 2017; Malone, 2003; Schumm & Hanelt, 2021). In essence, this new organizational form is characterized by an online community aligning around a common goal, being responsible for governing, managing, and improving the organization, and using blockchain technology to coordinate their actions (Mini et al., 2021; Schirrmacher et al., 2021). The community members' knowledge and resources, which are shared, transferred, and co-created in a transparent manner using computer-mediated communication tools (i.e., Reddit or Discord), build their organization's backbone (Beck et al., 2018; Ziolkowski et al., 2020).

As the phenomenon itself, also research on DAOs is still emerging. While existing research highlights the potential application areas to revolutionize, for example, the healthcare industry (Mateus & Sarkar, 2023) or the financial

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industry (Bellavitis et al., 2023; Mini et al., 2021) and provides an overview of the different forms and shapes of DAOs (Hsieh et al., 2018; Wang et al., 2019; Ziegler & Welpe, 2022), it lacks a common view and conceptual clarity of DAOs (Hassan & De Filippi, 2021; Santana & Albareda, 2022). This issue is also acknowledged in ongoing debates ranging from unclear definitions of the terms decentralization and autonomy (Hassan & De Filippi, 2021; Santana & Albareda, 2022) to the conceptual differences of closely related concepts such as open-source software (OSS) (Hsieh et al., 2018; Schirrmacher et al., 2021). However, this unclarity is problematic for both practitioners and researchers. For practitioners, an unclear conceptualization can lead to the development of different understandings of DAOs, as we currently witness (Hassan & De Filippi, 2021), limiting the potential to implement and develop shared standards and benefit from common learnings and failures. These shared standards can be beneficial to educate the broader community on critical attributes of DAOs, playing a crucial role in preventing ongoing scams such as so-called rug pulls in which the founders or developer team present their project as a DAO to attract capital from investors and the broader community to then drain the organization from its funds. For researchers, an unclear conceptualization can hinder the application and development of theories in the DAO context (Hong et al., 2014) or, worse, lead to a misuse and fragmentation of research on the phenomenon. Additionally, it can limit the application of DAOs as empirical cases in the broader development of research on more decentralized forms of organizing that currently lack large-scale empirical investigations. Another crucial aspect researchers face that is often missing when studying emerging phenomena, such as DAOs, is the role of context, referring to the set of factors surrounding the phenomenon (Avgerou, 2019; Davison & Martinsons, 2016). While IS literature has put some focus on the contextual features of DAOs from a governance perspective (e.g., Beck et al., 2018; Mini et al., 2021; Ziolkowski et al., 2020), it has mostly neglected other potential contextual features, such as the underlying beliefs and views of their members, leaving an incomplete view of DAOs and their context, up to date (Santana & Albareda, 2022). To address these issues, it is argued that more empirical and field research on DAOs can provide further clarity about the concept of DAOs (Santana & Albareda, 2022) with several researchers calling for further research in this area (i.e., Beck et al., 2018; Mini et al., 2021; Ziolkowski et al., 2020). Thereby, especially the perspective of and description by DAO members, providing DAOs with their knowledge and resources, can lead to new insights into this complex phenomenon (Beck et al., 2017; Santana & Albareda, 2022). While exploring the members' understanding of the phenomenon can be quite challenging due to their operations under a pseudonym, tapping into this source of knowledge can provide researchers with a more detailed perspective on the phenomenon (Beck et al., 2017). Thus, our research draws on a two-staged research procedure adapting elements of a netnographic approach to guide our data collection phase due to its systematic guidelines for the identification of a suitable online forum, familiarization with the given online forum, and its methodological focus on online communities using transparent computer-mediated communication in written text form where the real identities of the involved actors are unknown (Kozinets, 2002). For the data analysis phase, we adapted a structural topic modeling approach to manage the vast amount of digital trace data produced by the online communities (Schmiedel et al., 2019). This leads to the following research questions:

RQ 1 *How are DAOs described by their members?*

RQ 2 *What are the contextual features that shape DAOs in contrast to closely related concepts?*

Our study contributes to research and practice in several ways. From a research perspective, our work provides further clarity on the concept of DAOs through an empirical investigation of the perspective of their members, introducing a so-far unexplored perspective. Additionally, our findings highlight the contextual features surrounding DAOs, helping to set DAOs apart from closely related concepts such as OSS communities. From a practice perspective, this work helps to open the silos DAO members usually operate in and provides an overview to practitioners of the general ecosystem of DAOs.

The remainder of this paper is structured as follows. We briefly outline the current state of research regarding the topic of DAOs and OSS communities by highlighting already-known contextual features, such as their organizational characteristics and the studied governance models. In the next step, we give a detailed description of our applied two-staged research procedure, adapting elements of a netnographic approach in combination with structural topic modeling. Finally, we discuss our results by showing additional contextual features of DAOs and developing future research directions.

Theoretical background

Decentralized autonomous organizations

DAOs are a relatively new and emerging topic in IS research. While the origin of the term itself dates back to the 1990s, referring to multi-agent systems in an Internet of Things (IoT) setting or counter-movements against globalization,

explications of DAOs, which are closer to our current understanding, appeared a few years after the introduction of Bitcoin (Hassan & De Filippi, 2021). In general, DAOs are characterized by globally distributed members with a shared purpose (such as product building and management, investing, or community building (Ziegler & Welpe, 2022)) that use blockchain technology to virtually collaborate in a non-hierarchical, decentralized fashion with the help of a digital token (Beck et al., 2018; Mini et al., 2021; Wang et al., 2019). Thereby, DAOs use a public blockchain, such as Ethereum, to deploy smart contracts that entail the business logic of the offered product or service, as well as smart contracts that entail the governance rules determining how the underlying rules can be changed and when a decision is made (Wang et al., 2019; Zhao et al., 2022). Although most DAOs follow the purpose of building and managing a product or service, DAOs can be clustered along three main characteristics (community, governance, and treasury) with several sub-characteristics, allowing for a more fine-grained understanding of the concept itself (Ziegler & Welpe, 2022).

The characteristic community provides, for example, further information about the access requirements, the hierarchical structure, contribution rewards, or the purpose of the DAO (Ziegler & Welpe, 2022). Most DAOs have open access communities through open discussion forums or Discord servers, a flatter hierarchical structure, reward their contributors in the form of governance tokens or other tokens, and focus on product building and management (Hsieh et al., 2018; Mini et al., 2021; Schirrmacher et al., 2021; Wang et al., 2019; Ziegler & Welpe, 2022).

The characteristic governance, which is also widely discussed in IS research (i.e., Beck et al., 2018; Mini et al., 2021; Ziolkowski et al., 2020), provides more information on the type of governance token used, requirements for the governance process, and voting. For example, all studied DAOs in this recent work require the ownership of the governance token to participate in the process, while also the voting power is determined by the number of tokens a community member holds (Ziegler & Welpe, 2022). Furthermore, most DAOs require a combination of an off-chain (referring to the not directly encoded rules and processes (Reijers et al., 2021)) and on-chain (referring to the directly encoded rules and processes (Reijers et al., 2021)) governance process. However, as one might imagine, research has highlighted the issue that these governance processes, in their current form, still heavily rely on some centralized form of authority, from either a monetary perspective (i.e., so-called whales, which represent wealthy individuals holding large numbers of the governance token and thereby accumulating a large part of the decision power) or a knowledge perspective (i.e., developer teams accumulating the whole technological knowledge of the organization around them) (Beck et al., 2018; Ziolkowski et al., 2020).

The characteristic treasury provides further information on various factors surrounding the financing of the respective DAO. For example, most DAOs receive an income through their offered services, while others gain money by token sales or investment returns (Ziegler & Welpe, 2022). Additionally, it highlights the initial setup of the financing through, for example, token sales or airdrops (Ziegler & Welpe, 2022).

Open-source-software communities

Classifying DAOs in relation to other types of online communities requires a more nuanced understanding of OSS communities, as some early research suggests a relationship between the concept of OSS communities and DAOs (i.e., Hsieh et al., 2018; Schirrmacher et al., 2021), and most current DAO communities share the purpose or goal of product development and management (Ziegler & Welpe, 2022).

The well-established and mature research area of OSS in IS and management research can be mainly grouped into the research streams organizational aspects, governance, and motivational aspects (Von Krogh & Von Hippel, 2006). This research provides a deep understanding of the different roles of members within OSS communities, including developers, a large group of participants in online discussion forums, and so-called "lurkers" who are passive listeners and observers (Nonnecke & Preece, 2000). In addition, it highlights the complexity of governance of the product, including social norms, foundations, brands, and other mechanisms with several core governance principles such as independence, pluralism, representation, and decentralized decision-making (i.e., O'Mahony, 2003; O'Mahony, 2007). Thereby, motivational aspects are at the core of OSS communities since the project's success depends on the community's active contributions (Von Krogh & Von Hippel, 2006). Considering, for example, reasons for contributions to the overall success of the project, several studies highlight the central role of intrinsic motivation, such as ideology or reputation within the community (i.e., Choi et al., 2015; Stewart & Gosain, 2006; Von Krogh et al., 2012) as well as extrinsic motivation, such as paid participation (i.e., Roberts et al., 2006).

Research methodology

Our research aims at providing further clarity on the concept of DAOs. To address this issue, recent work has argued that more empirical investigations are missing, especially from the perspective of their members (Beck et al., 2017; Santana & Albareda, 2022). Because their members are organized and coordinated virtually in a transparent manner using computer-mediated communication tools such as Reddit or Discord (Beck et al., 2018; Ziolkowski et al.,

2020), they provide researchers with rich, timely, and continuous naturalistic digital trace data (Kozinets, 2015). Therefore, our research draws on two different but complementary approaches, which we combined within a two-staged research procedure. For the data collection phase and identification of a suitable online forum, we adapted a netnographic approach developed to study communities in a virtual environment where the existing data is often text-based (Kozinets, 2015). This adaptation for the data collection is not uncommon within IS research (e.g., Tarafdar & Kajal Ray, 2021) because it provides researchers with a systematic and coherent approach for the identification of suitable online forums and familiarization with the specific community's characteristics and language (Kozinets, 2002). Because in online settings, the amount of digital trace data can easily number in the hundreds of thousands of documents reaching limitations in the information processing capacities of individuals quickly (Berente et al., 2019), we followed a structural topic modeling approach (Roberts et al., 2016) that was adapted for organizational research (Schmiedel et al., 2019) for the data analysis.

Data collection

Our data collection is guided by the methodology suggested by Kozinets (2015). For the entrée, especially two steps are crucial: (1) having a specific research question and (2) identifying a suitable online forum to conduct the study. Since our study is guided by the questions to understand how DAOs are perceived by their members and the contextual features shaping this phenomenon, only the identification of an appropriate online forum is missing. We investigated several discussion forums about DAOs, including Reddit, Medium, and Discord. Although Medium and Discord provide deep insights into the DAO cosmos, they were excluded for the following reasons. While Medium is relevant for our topic, has a high traffic of DAO-related postings, and therefore contains rich and detailed data, it lacks active discussions between members related to these postings and thus rather represents the sole authors' opinion. Discord, on the other hand, has several active ongoing discussions about DAOs but suffers from the limitation of data accessibility, which might introduce biases in our results. In contrast to this, Reddit was identified as a rich source of discourse about DAO-related discussions, which offers, at the same time, the possibility to extract data throughout the whole platform, not limiting our scope to a few specific subreddits or groups. Therefore, we decided to choose Reddit as the most suitable online forum for DAO-related discussions, which is congruent with existing research in this area (DuPont, 2017). Reddit, in general, is a website founded in 2005 that comprises user-generated content, including photos, videos, links, text-based posts, and content discussion. Reddit describes itself as the home to thousands of communities, endless conversations, and authentic human connections (Reddit, 2022).

Another crucial aspect, before conducting the data collection and analysis, is to get familiar with the community's characteristics (i.e., interests and language) (Kozinets, 2002). Especially the language used in relation to cryptocurrencies and DAOs was very specific on Reddit and included many different abbreviations. To increase our understanding of the discussions, we, therefore, decided to check for the most common abbreviations and got familiar with them (i.e., BUIDL = purposeful misspelling of "build" for ironic meaning, ELI5 = explain it like I am 5, DEX = decentralized exchange).

Having identified a suitable forum for DAO-related discussions and gotten familiar with their specific language, our next step included the collection of data. Due to the coverage of the TheDAO hack in academic literature by DuPont (2017), our data collection started after the 1st of July 2017 and ended on the 22nd of February 2022. To collect data from Reddit, we used the PushShift API, a platform that collects, analyses, and archives Reddit data publicly since 2015 (Baumgartner et al., 2020). It was chosen due to the provision of extended functionality to query and retrieve data with the PushShift API, as well as the size limitations of retrievals with the official Reddit API (Baumgartner et al., 2020; Safadi et al., 2021; Weber et al., 2022). For our search, we used the string "DAO" since results with, for example, "decentralized autonomous organization" were too limited, leading to a total of 114,514 submissions and 292,220 comments on Reddit.

Data preparation and analysis

Our data was analyzed using the structural topic modeling (STM) approach by Roberts et al. (2016), which was adapted by Schmiedel et al. (2019) for organizational research. We chose the topic modeling approach for the discovery of frequently discussed topics in the Reddit data due to the sheer amount of data, which is not an uncommon approach when conducting a netnography (Kozinets, 2015). Topic models automatically infer latent topics from texts and allow researchers to discover topics from the data instead of assuming them beforehand and classifying texts accordingly (Roberts et al., 2014). This method can be classified as a form of unsupervised machine learning because it does not require us to label a set of training documents with a set of predefined topics to train the model (Roberts et al., 2014). An innovation of STM compared to older topic modeling methods, most notably Latent Dirichlet Allocation (Blei et al., 2003), is the possibility to incorporate metadata of the texts to guide the modeling process. In our case, the subreddit of each contribution is included since this is likely to influence the specific topics contained in each text.

After downloading the data, we observed that many irrelevant submissions and comments were present. The main

reason for this was the inclusion of subreddits not relevant to decentralized autonomous organizations that used the abbreviation DAO in different contexts (mostly spiritual or fantasy video games and movies). Therefore, our first step included the removal of all irrelevant subreddits, which was done by manually verifying if the subreddit contained discussions relevant to our subject of interest. This process was done quicker by removing all posts from subreddits with fewer than 40 mentions of the term "DAO" first, which amounted to about 10% of the total data. Following this, all posts marked as "removed" (usually spam posts that were removed by Reddit moderators) were also removed from the data.

The data was cleaned further by removing emoji symbols, URLs, and HTML tags. Duplicate posts were removed, as well as posts containing certain blacklisted words (related to irrelevant contexts for the term "DAO") and blacklisted patterns frequently seen in spam posts (Schmiedel et al., 2019). Very short comments (fewer than three words) were removed as well since these do not contain much information and might add noise to the model. Lastly, non-English posts were filtered out using Python packages *langid* (Lui & Baldwin, 2012) and *langdetect* (Danilak, 2015). After all filtering steps, 73,064 total posts remained in the dataset.

We used the *stm* R Package (Roberts et al., 2019) to create the topic models and further prepare the text beforehand with steps such as removing stopwords, stemming words (reducing words to their root form), and tokenizing (splitting into individual terms) the texts. Since topic modeling requires us to specify the number of topics beforehand, and no single correct number of topics can be determined (Roberts et al., 2019), we created multiple topic models to see the effect of choosing different numbers of topics. Roberts et al. (2019) recommend using various metrics about the quality of a topic model to guide the selection process: semantic coherence, which is high when frequent words of a topic often appear together in the texts (Mimno et al., 2011), and exclusivity, which measures how distinct the top words are to a topic (Bischof & Airoldi, 2012).

Figure 1 shows both metrics for various topic models created for the Reddit data. Both metrics usually present a trade-off (Roberts et al., 2016), which can also be seen in Fig. 1, where most solutions are not strictly better than others. After inspecting all created models for the interpretability of their topics (Schmiedel et al., 2019), we chose the model with 25 topics as the best categorization of the data. Increasing the number of topics led to a larger number of confusing topics without adding much new information. According to the exclusivity and semantic coherence metrics, this model is also strictly better than the model with 20 topics and provides a reasonable middle ground between both metrics.

In addition, to interpret and label these topics, we used the *stmcorrviz* package (Coppola et al., 2016), which highlights the most important words per topic and provides exemplary review text for each topic, as well as a categorization of

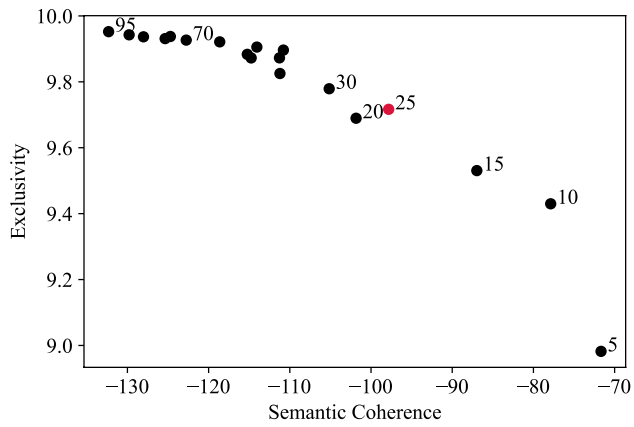

Fig. 1 Exclusivity and semantic coherence for different numbers of topics

related topics. More exemplary reviews were retrieved using the *findthoughts* function of the *stm* package (Roberts et al., 2019). We labelled the topics by first looking at the most highly associated terms, meaning the words with the highest probability of appearing within the topic. Considering, for example, topic nine, these included *token*, *vote*, *fund*, *will*, *holder*, *dao*, and *propos*. Additionally, the stm package provided us with an exemplary review text, such as "*[…] These governors, via decentralized aggregation mechanisms such as a DAO, will have the power to put forward measures for our quarterly governance votes. These measures will still be voted on by our current governance platform, which will not change. […].*" Since the highly associated terms and the review text revolved around voting, we labelled the topic "Voting." To ensure a high reliability of our results, the topics were labelled by two researchers independently (see Appendix Table 2 for a complete list).

We proceeded by identifying the relevant topics for our research question. Topics not relevant to our research question were excluded (Schmiedel et al., 2019). For example, topic seven was excluded due to its high prevalence of project advertisements with no contribution to our research question (example text: "*Hey! Check out this amazing NFT project […]. NFT holders also get passive income through a Community DAO Wallet!*"). Finally, this led to 16 relevant topics about DAOs, which essentially represented our 2nd-order themes. We then further investigated and distilled them to 2nd-order "aggregated dimensions" (see Fig. 2) (Gioia et al., 2013) (Table 1).

Results

Based on our analysis, we derived 16 relevant topics which show insights into how DAO members describe the phenomenon and their surrounding contextual features, providing a detailed answer to our research questions. First,

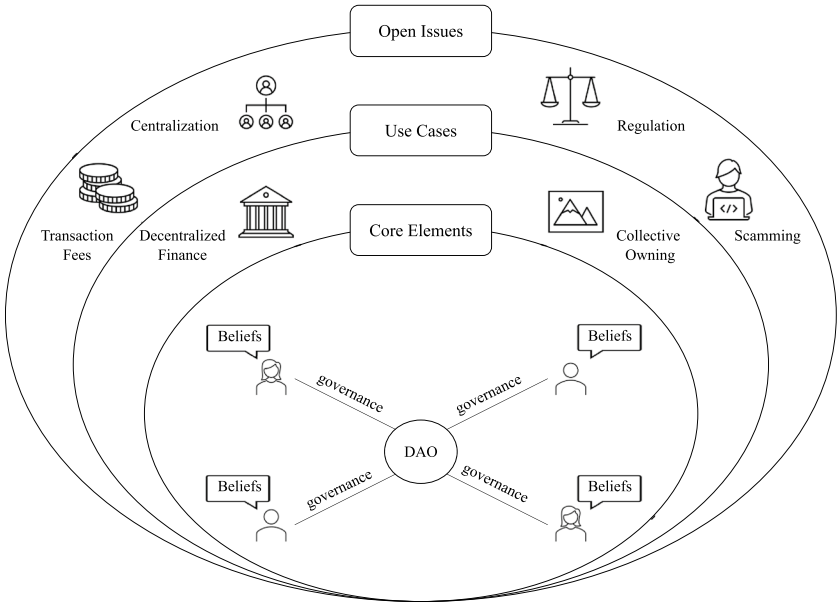

Fig. 2 Overview of DAOs from the perspective of their members

we show the core elements of DAOs from the perspective of the online communities by presenting definitory aspects, providing an answer to our first research question. Second, by identifying the governance structure and introducing the online communities' beliefs and views, we provide first contextual features of DAOs. Having understood the core concept of DAOs, we present additional contextual features such as the most named use cases in the context of DAOs, namely decentralized finance (DeFi) and collective owning, as well as existing open issues, such as possible re-entrance of centralization, the unclear regulation around DAOs, scamming, and transaction fees. For better illustration and transparency, we included exemplary quotes from our dataset. Those quotes were chosen manually by the authors based on unanimous decision from a list of the 100 most representative posts for each topic (based on their probability of belonging to that topic).

Core elements

The community members provide a detailed understanding of DAOs, and their essential design features and structures, giving an answer to our first research question. One recurring theme is the introduction of a definition for DAOs, which was in many cases derived from Wikipedia, as one member describes it as follows:

Table 1 Final topic labels

Topic label	Highly associated terms	Topic ID
Scamming	time, api, treasuri, olympus, scam, ohm, wonderland	1
Influential Personalities	dao, fund, decentralis, investor, vitalik, live, mention	2
Miner Centralization	network, block, node, miner, central, mine, valid	3
TheDAO Emotions	just, peopl, dont, money, get, say, even	5
Collective Owning	world, can, buy, real, one, ownership, creat	8
Voting	token, vote, fund, will, holder, dao, propos	9
Hard Fork	ethereum, fork, eth, etc, hack, hard, chain	10
Refund	address, account, use, post, send, can, withdraw	13
Joining	dao, like, look, think, get, see, work	15
DeFi	stake, liquid, fee, pool, defi, protocol, token	16
Regulation	ico, secur, legal, articl, million, rais, sec	17
Code is Law	contract, code, smart, bug, attack, law, dao	18
Structure	communiti, govern, decis, member, control, discuss, structur	20
Fees	ethereum, develop, dapp, will, blockchain, smart, use	21
Beliefs	think, will, point, way, peopl, problem, one	23
Differentiation	first, make, alreadi, dao, run, sinc, mean	25

> […] A decentralised autonomous organisation, sometimes called a decentralised autonomous corporation, is an organisation represented by rules encoded as a computer program that is transparent, controlled by the organisation members and not influenced by a central government. (Topic 20)

Thereby, the communities' members attach great importance to different aspects of a DAO, which are also found in various other definitions: encoded rules in the form of smart contracts that are transparent and controlled by the members without a central instance of control as well as autonomous operational activity through code. Especially on the aspect of decentralization, the members put a major focus on arguing that the executive decision power should not rest with one single person.

Governance (Topic: 9, 15, 20, 25)

To guarantee this ongoing decentralization, the construction of a proper governance model takes center stage within DAOs. Within these models, the governing body of the organization can vote on proposals in their respective governance forum and make decisions using tokens that a community member compared, for better illustration, to shares in public companies:

> […] a vote is brought forth before the governing body which cannot be manipulated. Similar to the relationship of class A and C shares. (Topic 20)

The members argue that a usual governance process starts with the submission of a so-called improvement proposal by one or more community members to the respective governance forum. These proposals are then open for discussion with the whole governing body of the DAO, as well as other interested stakeholders. Depending on the type and importance of the proposal, this step is usually paired with a specific timeframe to limit the discussion time and avoid an infinite discussion, as one user argues:

> […] Usually proposals are a few days up to a few weeks. It depends on the proposal and the amount of discussion needed […] (Topic 9)

The next step involves deciding whether the suggested proposal should be implemented. Therefore, DAOs make use of a concept called token voting, meaning that community members, who are interested in the governance process of the DAO, are holding the respective governance token of the DAO, enabling them to vote on proposals. The decision power of an individual can be determined in various ways. The most common one is argued to be "one token, one vote," meaning that each token represents one voting power for a decision. Put simply, the more tokens an individual holds, the larger its decision power.

> Normally in a DAO, like say COMPOUND's DAO anyone who hold COMP tokens can vote. But the vote is weighted based on how much COMP you hold. In my opinion it is a plutocracy. Whales win. (Topic 9)

As with every existing governance model, also this one is not free of limitations. Within the DAO communities, it is often viewed critically. For example, the user above mentions the risk of a plutocracy when conducting token-weighted voting, meaning that an organization is controlled by wealthy individuals or groups. In the above-cited comment, the user criticized by the DAO suggested governance model, which would require around $60,000 to participate. However, interestingly, discussions that criticize the existing governance models within DAOs often use the comparison between DAOs and public companies. They especially argue that token-based governance with a fully democratized decision-making process has several shortcomings, such as the inability to make quick decisions. Furthermore, they argue that, in contrast, public companies' decisions are made by strong executive bodies to address this issue. Therefore, users argue that a possible lack of executives within DAOs could create a dysfunctional governance model:

> […] My rule of thumb: if your token governance design, or your DAO, does not have executives, i.e. people who can take binding actions first, and only be punished after the fact, then you've designed a dysfunctional governance system. (Topic 20)

Nevertheless, this criticism often refers to simple governance models where everything is voted on. It excludes token-based governance models that go beyond this token-weighted voting, for example, the concepts of Colony, DAOstack, Aragon, or quadratic voting. Especially Colony's governance model is highlighted in this context since it

> […] is designed to allow people to make empowered binding decisions first, and then to later suffer the consequences if they angered the group. (Topic 20)

Thereby, Colony aims at solving the issues with token-weighted voting in a different way by weighting the votes based on so-called "reputation," which is earned by contributors when getting paid by the DAO in its native token. Over time, the reputation decays, making it possible for new contributors to add value and have a say in the governance process.

Contrasting beliefs (Topic: 5, 10, 13, 18, 23)

Another crucial core element, as mentioned above, are the members themselves, holding a unique role within a DAO by providing their knowledge and resources and being responsible for governing the organization. Since these members are operating in many cases under a pseudonym, understanding

their underlying beliefs and views is of utmost importance and shapes the context of DAOs.

In general, we could identify that there is an ongoing rivalry between competing blockchain ecosystems, such as Bitcoin and Ethereum, with both sides complaining about users talking disparagingly about other projects in the respective subreddits:

> Soooo you were annoyed by the small percentage of people from here who went to bitcoinmarkets and talked shit, and have decided to paint everybody here with the same brush and do the exact same thing those people did that annoyed you, to this subreddit? […] (Topic 5)

Interestingly, as presented by the user above, these posts are often argued to represent just a small amount within a specific DAO community. Within these discussions, community members often put a large emphasis on providing arguments for claims being made to differentiate those claims from what they call "trolls," which a user explains as someone making up things to suit their narrative.

While these claims mainly referred to discussions between different blockchain ecosystems, the TheDAO hack and the followed hard fork of the Ethereum blockchain (the split of the Ethereum blockchain) introduced another controversy within the Ethereum community itself. The split emerged from the handling of the TheDAO hack and is based on two different assumptions and views and lasts until today. On the one side of these discussions are community members who view the written code as law and, therefore, argued against a rollback of the blockchain. On the other side are community members who argue to fork the blockchain and therefore roll it back, with the consensus of the majority of miners and other stakeholders because:

> […] Ethereum was so young that the hack (and the amount of stolen ETH from the smart contract) threatened the entire project […]. (Topic 10)

Community members supporting the ethos of "code is law" are now operating the Ethereum Classic blockchain (ETC). Their arguments are based on the belief that everything which is encoded in the contract should be considered a law because investors in TheDAO "signed that contract" and that a blockchain is immutable. This also includes the circumstances in which the contract is poorly coded:

> No, they kept the promise that contract was law. The DAO wasn't robbed, it was poorly coded and someone exploited that. If contract is law then the exploiter was in the right. (Topic 18)

Contrary to this, community members who support the rollback of the Ethereum blockchain are operating the Ethereum blockchain (ETH), which is known to most people today. Their arguments are based on the belief that TheDAO was indeed a hack because, although having a bug, it was not the intention of TheDAO to be hackable. Therefore, a rollback with the consensus of the miners was acceptable. Thereby, many users argue that they did not strictly follow the ideals of decentralization or immutability of code but that they weighed the pros and cons of the exploit and made a decision based on social consensus, often referred to today as "code is constitution":

> […] Those of us who supported the pro-fork side over the DAO case may actually move to the anti-fork side in the future. It all depends on the kinds of circumstances we're presented with. […] No sane rational person will ever accept a principle on absolute terms. People who signed up for decentralization or immutable code at all costs are part of an ideological minority with a deeply flawed world view. […] (Topic 23)

Use cases

Having understood the core elements of DAOs, members also discuss different use cases for this concept. The two main discussions revolve around the application of DAOs in the DeFi sector and the collective ownership of assets.

Decentralized finance (Topic: 16)

Within the DeFi sector, DAOs are used as a governance concept to govern and manage the, in many cases, highly automated and autonomous financial products, including decentralized exchanges, allowing users to trade their cryptocurrencies (i.e., Curve finance), or liquidity protocols, allowing the user to earn interest and borrow assets (i.e., Aave).

Collective owning (Topic: 8)

In addition, the collective ownership of assets is discussed within the virtual space, meaning the acquisition of, for example, non-fungible tokens (NFTs), as well as the non-virtual space, for example, the acquisition of real estate or art assets. The two projects which are discussed most are a DAO called ConstitutionDAO, which bid $42 million for a copy of the US constitution at Sotheby's auction, and a DAO called PleasrDAO, which acquired the Wu Tang Clan album "Once Upon A Time in Shaolin" for $4 million:

> […] PleasrDAO, the collective of decentralized finance leaders, NFT collectors and digital artists, announced itself as the new owner of Wu Tang Clan's one-of-a-kind album "Once Upon A Time In Shaolin." The 74 members of the decentralized autonomous organization (DAO) now share collective ownership of the unique projects. […] (Topic 8)

Open issues

As with every emerging phenomenon and a largely unregulated market, DAOs are also surrounded by risks and unaddressed issues that have to be solved, including scams, a re-entrance of centralization through influential personalities or miners, transaction costs, and unclear regulation. These issues mostly emerge through the specific contextual features of DAOs, such as their given governance structure, beliefs and views, and application areas.

Scams (Topic 1)

A significant issue within DAOs is the pseudonymity of actors introducing a risk in the DAO. Consequently, there are active discussions and warnings around current scams, with, for example, one user claiming that 75% out of 50 DAOs he/she researched were scams. Those scams are often advertised by promising extremely high returns when investing in the respective DAO:

> My buddy invested in Olympus DAO. He says the APY [annual percentage yield] is over 7000%. Is he being scammed? 7000% APY seems astronomically high. Immediately threw up red flags for me. Is this project legit or did my buddy get scammed? (Topic 1)

Centralization (Topic 2, 3)

In addition, although DAOs are argued to be completely decentralized, there still exist influential personalities or developer teams that are actively shaping a DAO and its direction, introducing some form of centralization around this person or team. Thereby, members actively rely on a person's or team's reputation in a positive or negative way. For example, reputation is used to argue for the solidity of a project because a well-known person within the ecosystem is backing the project. However, on the contrary, these influential personalities can also be associated in a negative way when, for example, deciding who is responsible for something:

> He marketed the DAO extensively. From the SEC's perspective for example, he is responsible for it. He said he reviewed the code, and he has now worked on the code for a HF to bail out DAO investors. You can't separate Vitalik from the DAO. (Topic 2)

This possible issue of a re-entrance of centralization in the organization is also found in the centralization of miners, who are responsible for verifying the transaction added to the blockchain. This is especially relevant since votes are conducted on the underlying blockchain and verified by the miners (in the case of the proof-of-work consensus mechanism). Centralization, in this case, is argued to potentially limit the DAO since needed upgrades could be blocked, the consensus mechanism could be changed to a more centralized one, or unwanted changes could be pushed through. However, it is also argued that centralization and decentralization are instead a continuum, with both having their trade-offs. As one member states in a discussion about whether the EOS project is rather an expensive server instead of a decentralized blockchain:

> It depends on what you believe in, would you rather have 3-4 mining pool controlling all the hash without you playing a single role into it but being deceived that it is all decentralized or a semi-centralized network in which you choose your own semi-centralization points so they can recover in attacks like DAO and doesn't have to go through forks and ensure higher scalability and no fees? (Topic 3)

Transaction fees (Topic 21)

Depending on the chosen blockchain, DAOs can also face another issue which was mentioned by the user above, high transaction fees. This becomes especially an issue when votes are conducted on the blockchain, making it unattractive for DAO members to vote due to high fees for executing the transaction and, therefore, limiting participation, which introduces a risk to the DAO. As a possible solution to this issue, several DAO members introduced different blockchain protocols which are addressing the issue of high transaction costs and scalability, including so-called Layer 2 (a second protocol that runs on top of an existing blockchain) or completely different blockchains such as EOS or Cardano.

Regulation (Topic 17)

Finally, DAO members discuss the still unclear regulatory status of DAOs and introduce different regulatory approaches based on different jurisdictions. For example, a central topic of discussion revolves around the introduction of new legislation within Wyoming (USA) which allows a DAO to be registered as a limited liability company (LLC):

> […] Wyoming's DAO-focused "Bill 38" has been approved by the state senate. The bill would allow decentralized autonomous organizations (DAOs) to gain legal company status in the state of Wyoming. DAOs would be able to register themselves as limited liability companies (LLCs) and more easily engage in contracts and legal agreements. […] (Topic 17)

Besides Wyoming, the members also discuss other jurisdictions having "FinTech sandboxes," such as the UK,

Singapore, Liechtenstein, and Switzerland. Even the EU is mentioned to work on possible legislation. This is seen as extremely important since many initial coin offerings are argued to be classified as a security offering in the USA since TheDAO in 2016:

> […] the SEC issued a Report of Investigation under Section 21(a) of the Securities Exchange Act of 1934 describing an SEC investigation of The DAO, […] determining that DAO Tokens were securities. The Commission stressed that those who offer and sell securities in the U.S. are required to comply with federal securities laws. […] (Topic 17)

Discussion

The aim of our research was twofold. First, our research aimed at extending our understanding of the emerging phenomenon of DAOs. Second, our study aimed at uncovering the surrounding contextual features of DAOs that shape and distinguish them from other closely related concepts. By introducing a previously unexamined perspective, namely by the members of the underlying DAO communities (Beck et al., 2017), our results provide a more holistic picture of the phenomenon. Our work contributes to research in two significant ways based on our findings. First, we are closing a research gap by Beck et al. (2017) and Santana and Albareda (2022), who identified the necessity to study blockchain developers and communities further, representing a vital resource and knowledge base for DAOs. With our two-staged research approach, we could derive further insights about DAOs from the perspective of their members and thereby gain a deeper and more holistic understanding of the phenomenon. We show that their descriptions of DAOs are similar to current research and even extend existing research by uncovering so far unstudied governance processes. These processes differ from existing ones regarding the determination of decision power and in which domain a community member holds the decision power. Second, our research provides further clarity on the concept of DAOs by identifying additional contextual features, such as their underlying beliefs and values, and use cases. These contextual features are especially relevant for developing and adapting theories in the context of DAOs and can be helpful to set DAOs apart from closely related concepts.

Contrary to the criticism towards their members for being too optimistic (Beck et al., 2017), our findings highlight that their descriptions of DAOs align with current research on the topic and even extend these works in some regards. Considering, for example, the core elements of DAOs, it can be studied that both researchers and DAO members highlight characteristics such as encoded rules on a blockchain and a decentralized operation of the DAO by distributed members with a shared purpose and goal (Beck et al., 2018; Mini et al., 2021; Ziolkowski et al., 2020). To ensure an ongoing decentralization (Wang et al., 2019), we studied that both researchers and DAO members agree on the importance of a proper governance process within DAOs which is broadly compromised of (1) a suggestion of a proposal by a community member, (2) discussion about the proposal, and (3) voting on the implementation of the proposal by means of a governance token (Schirrmacher et al., 2021; Zhao et al., 2022). In this regard, the members' descriptions even extend the academic perspective by uncovering so far unstudied governance mechanisms. While research often argues that the decision power is determined by the number of tokens a voter holds (Mini et al., 2021; Schirrmacher et al., 2021), the DAO members provide examples of voting processes that go beyond the typical "one token, one vote" mechanism. This variation of processes includes the governance process by Colony, DAOstack, or quadratic voting. For example, Colony uses a reputation-based system, meaning that community members can contribute to the DAO and thereby get paid and earn "reputation" in the domain they contributed to. This reputation gives members the power to vote on decisions within the given domain (Colony, 2022). Additionally, we studied that DAO members were also able to identify the same occurring issues as researchers within their governance processes (i.e., influential personalities (Ziolkowski et al., 2020)).

In addition, as this comparison of the DAO members' descriptions and our current academic understanding of DAOs highlights again, there is a focus on structural aspects of DAOs' organizational structure (e.g., Hsieh et al., 2018; Wang et al., 2019) or the applied governance processes (e.g., Beck et al., 2018; Mini et al., 2021; Ziolkowski et al., 2020). However, research emphasizes the necessity of knowing the contextual features of new phenomena and cases (Avgerou, 2019; Davison & Martinsons, 2016), being crucial for contextual theorizing. By uncovering the contextual features of DAOs, such as their governance processes, underlying beliefs and views, use cases, and current open issues, our work highlights the early stage of research on DAOs and the phenomenon itself. If we compare our identified contextual features to, for example, OSS research that is sometimes indicated to be closely related to DAOs due to similar organizational structures (Hsieh et al., 2018; Schirrmacher et al., 2021) and the shared goal and purpose of product building and management (Ziegler & Welpe, 2022), we can further elucidate similarities and differences.

Our findings show that the governance process within DAOs often follows a similar logic starting with identifying an issue or suggestion for improvement brought forward to the DAO members in the form of so-called improvement proposals within the respective governance forum. This proposal is then open for discussion and can be attached with a timeframe to hinder infinite discussions. The decision on whether or not the proposal should be implemented is then decided upon with token voting, while the determination of voting power can range from one token, one vote to reputation-based approaches. Contrary to this, governance processes within OSS communities are often less systematic. They can follow different iterations through the overall steps of identifying an issue, developing a solution, evaluating the solution, and announcing the changes (Eseryel et al., 2020). Additionally, while OSS communities also lean towards a more decentralized decision-making process (O'Mahony, 2007), decisions are often made by core developer teams (Eseryel et al., 2020) compared to DAOs that try to involve all token holders within the governance process.

Regarding the underlying beliefs and views, our work highlights the two contradicting paradigms of "code is law" and "code is constitution," emerging from different views on how, or even if, code on the blockchain should be changed. Research on OSS, on the other hand, can provide already sophisticated and deep insights into the motivational structure and underlying ideologies of members within OSS communities (i.e., Choi et al., 2015; Stewart & Gosain, 2006; Von Krogh et al., 2012). For example, Stewart and Gosain (2006) show in their study the great value of OSS beliefs, such as Code Quality, Software Freedom, Information Freedom, Bug Fixing, Practicality, and Status Attainment, as well as OSS values, such as Sharing, Technical Knowledge, Learning, Cooperation, and Reputation, on the effectiveness within OSS developer teams.

Considering the next contextual factor, use cases, we show that although the primary goal of DAOs is still product building and management, its use cases can include DeFi applications and collective ownership of digital and physical assets. In contrast, OSS use cases are often limited to software development (AlMarzouq et al., 2005).

Last, DAOs are currently surrounded by several open issues such as high transaction fees that can limit participation within the governance process, centralization in the form of miner centralization and influential personalities that can determine the direction of a DAO, scams, as well as unclear, but emerging, regulation. Contrary to this, primary issues within OSS communities are the sustainability of the project due to a lack of active participation of its members as well as the scaling of such communities (Gamalielsson & Lundell, 2014; Martínez-Torres & Díaz-Fernández, 2014).

As this exemplary comparison based on our identified contextual features highlights, DAOs share some similarities with OSS communities but are currently surrounded by different issues that they have to overcome first. However, it is likely that issues known in OSS projects, such as a lack of participation and the difficulty of sustaining such a project, will also affect DAOs.

Limitations and future research

As with every study, also this one is not free of limitations. First and foremost, we want to emphasize that there are additional communication tools in which discussions about DAOs happen, apart from Reddit. Although Reddit is a very active discussion forum with DAO-related discourses (DuPont, 2017), the discourse shifts to the respective governance forums and Discord servers. For example, discussions revolving around MakerDAO are mainly happening now within their governance forum: https://forum.makerdao.com. This means that our results only represent the description of DAO members on this platform and are not necessarily applicable to other platforms, such as Discord or the governance forums. Second, although we compared the perspective of DAO members with a research perspective, we want to emphasize that our results still represent the opinions and views of pseudonymous users and that neither their identities nor their statements and intentions can be verified.

Therefore, future research should use our given approach to study additional communication tools with DAO-related discussions to identify commonalities and differences between the given tools. Future research should also test our findings regarding their members' underlying beliefs and views. Thereby, it would be especially interesting to see if the provided opinions represent the general DAO members' beliefs and which ideological assumptions and demographics are underlying the community members.

Additionally, our comparison of the identified contextual characteristics of DAOs and OSS projects has also opened up further research opportunities. Future research could use our identified contextual characteristics of DAOs to test and further develop existing governance theories from OSS research. It would be interesting to see how features such as a stronger focus on internal capital and treasury management influence the given theory. This would contribute to a better understanding of DAOs and help to set them apart from the concept of, for example, OSS.

Furthermore, future research should incorporate different perspectives on this emerging phenomenon of DAOs. As shown by our exemplified comparison of DAOs and OSS, especially the underlying beliefs and values are

much better known in the context of OSS research (i.e., Choi et al., 2015; Stewart & Gosain, 2006; Von Krogh et al., 2012), while we could, so-far, identify the beliefs on code immutability in the context of DAOs. Therefore, future research should focus on identifying additional beliefs and ideologies of DAO members. This can be especially interesting since some OSS values, such as the aspect of reputation, can also be found in the context of DAOs, considering, for example, the identified issue of centralization around reputable personalities or teams. A more detailed understanding of DAOs in this regard can also help to develop conceptual clarity further.

Conclusion

Our research was motivated by the unclear conceptualization of DAOs in academic literature and practice (Hassan & De Filippi, 2021), which represents a significant problem for setting DAOs apart from closely related concepts. To address this issue, our study followed a two-staged research procedure to enhance our current understanding of DAOs and elucidate contextual features surrounding this phenomenon. By providing further clarity on the concept and exemplarily contrasting DAOs with OSS, we have set a basis for a more sophisticated understanding of this phenomenon and future theorizing in this context.

Appendix

Table 2 Complete topics list

Topic ID	Topic label	Highly associated terms
1	Scamming	time, api, treasuri, olympus, scam, ohm, wonderland
2	Influential Personalities	dao, fund, decentralis, investor, vitalik, live, mention
3	Miner Centralization	network, block, node, miner, central, mine, valid
4	*Topic indeterminate*	valu, will, price, coin, market, asset, token
5	TheDAO Emotions	just, peopl, dont, money, get, say, even
6	*Topic indeterminate*	will, communiti, hoge, list, week, team, updat
7	*Topic indeterminate*	nft, game, nfts, will, launch, communiti, dao
8	Collective Owning	world, can, buy, real, one, ownership, creat
9	Voting	token, vote, fund, will, holder, dao, propos
10	Hard Fork	ethereum, fork, eth, etc, hack, hard, chain
11	*Topic indeterminate*	wallet, dao, stake, can, ckb, use, deposit
12	*Topic indeterminate*	eth, will, price, now, time, year, day
13	TheDAO Refund	address, account, use, post, send, can, withdraw
14	*Topic indeterminate*	project, crypto, invest, team, great, new, will
15	Joining	dao, like, look, think, get, see, work
16	DeFi	stake, liquid, fee, pool, defi, protocol, token
17	Regulation	ico, secur, legal, articl, million, rais, sec
18	Code is Law	contract, code, smart, bug, attack, law, dao
19	*Topic indeterminate*	dash, fund, propos, coin, monero, masternod, decr
20	Structure	communiti, govern, decis, member, control, discuss, structure
21	Fees	ethereum, develop, dapp, will, blockchain, smart, use
22	*Topic indeterminate*	decentr, platform, organ, will, manag, autonom, provid
23	Beliefs	think, will, point, way, peopl, problem, one
24	*Topic indeterminate*	blockchain, bitcoin, use, transact, can, cryptocurr, system
25	Differentiation	first, make, alreadi, dao, run, sinc, mean

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